LIFELINE — Approach Note

Group 24 | Generator seed: 24 | 50 synthetic cases: S (00–19), M (20–39), L (40–45), X (46–49)

1. From story to numbers

The generator turns six disaster-response stories into reproducible integer datasets. Cases 00–45 rotate through the six studies; sizes follow the assigned tier ranges, while scenario-specific overrides enforce each story's signature.

Case study	Data choices and story justification
1. Delta Flood	Path-connected, low-degree road networks with a few added links keep average degree between 2.2 and 3.2. At least 40% of locations have penalty ≥2; 3–5 disjoint closure waves are planted. Food/water crates are at least 55%. This reflects a flat delta of small villages joined by raised roads, where food and water matter most.
2. Mountain Earthquake	Road density stays sparse (average degree 2.0–2.6). Medicine/tool crates are at least 45%, and alarm priorities average at least 6. Sparse alternatives and bursts of urgent calls model a valley town with few roads after repeated aftershocks.
3. Coastal Cyclone	At least one quarter of locations are shelters, with at least three closure waves. Rescue windows deliberately overlap, and the radio log contains all 38 symbols. These choices represent storm-surge closures, short overlapping rescue windows and noisy radio traffic.
4. Urban Flash Flood	The road graph is denser (average degree ≥3.5), at least 10% of crates are rare equipment, and M/L alarm arrivals average 1.2–1.8 per tick with six or more priorities. This represents many alternative routes and calls arriving faster than responders can answer.
5. Landslide Corridor	A path-like graph with limited local chords keeps average degree ≤2.4 and maximum degree ≤4. At least three closure waves include a bridge whose removal isolates a target; HQ-to-farthest distance is at least floor(V/4). This models a single mountain highway where a collapse can cut off whole communities.
6. Outbreak Camp	Medicine/water crates form at least 55% of stock; at least half expire by hour 300. One radio symbol occupies at least 25% of the log. This models essential supplies that spoil and radio beacons that repeat the same messages.

2. Planted objects and claims

The generator constructs each object before writing the workbook, and the validator independently recomputes its validity. The road network starts connected; closure waves reference distinct road IDs. Three target IDs refer to distinct shelters/hospitals. The site tour is a permutation beginning at site 0; its closed Manhattan length is recomputed, and L/X tours use nearest-unvisited order. The sensor graph is a connected tree whose edges all touch the selected k-node cover. Each safety clause uses three distinct switches and is made true under a planted assignment. A selected subset of ration values is marked planted and its sum becomes rations_target.

For tier S, claims are computed rather than guessed: 0/1 knapsack DP for cargo value, weighted-interval DP for both window objectives, rod-cutting DP, Held–Karp exact closed-tour search, and exhaustive minimum vertex cover. The validator recomputes these values independently. Claims are absent from M/L/X cases.

3. The traps

Trap	What is generated and why it is hostile	Validator coverage
46 — Dock-order	1,500–3,000 crates are emitted in crate_id order. Value, expiry, serial and value/kg are monotone; weight is constant. A solver that assumes input order is useful receives no helpful ordering signal.	V2 crate ranges; V3 identifiers/serial uniqueness; V8 monotone-field count.
47 — Best-looking crate	Thirteen crates and capacity 20 kg are arranged so the 11 kg / value 100 crate has the best ratio and is chosen first, but leaves too little capacity for either 10 kg / value 90 crate. The two 90-value crates form an optimum worth 180; greedy obtains 100 (55.6%).	V2 weight/value/capacity; V8 recomputes ratio-greedy and exact optimum.
48 — Shortest looks cheapest	A 1,500–2,000-location chain gives a very long graph distance. A direct 0→2 road takes 60 minutes, while 0→1→2 takes two one-minute roads. The target at site 2 therefore has a one-road route that is far more expensive than the cheapest route.	V3 references; V4 connectivity; V8 graph-distance and minimum-hop versus cheapest-cost comparison.
49 — Near-miss	The 50,000–100,000-symbol log repeats the one-symbol block A. Five patterns are all-A prefixes; the 40-symbol pattern has 39 As followed by B, so it is absent while its 39-symbol prefix matches at essentially every start position.	V5 recomputes overlapping pattern counts; V8 checks repetition fraction and near-miss prefix frequency.

4. What will be hard

Case study	Two hardest decisions and the data property causing difficulty
Delta Flood	Cheapest route after closures: a connected graph can become disconnected when waves remove bridges. Cargo selection: indivisible crates and tight capacity make value/kg ordering unreliable.
Mountain Earthquake	Emergency queue: urgent alarms arrive in bursts and priorities tie. Safety planning: clauses share switches, so a locally attractive choice can violate several rules.
Coastal Cyclone	Rescue scheduling: overlapping windows create conflicts, while touching windows remain compatible. Radio comparison: noisy logs and near-matching reports make edit operations and overlapping matches important.
Urban Flash Flood	Route planning: dense alternatives increase candidate paths. Emergency queue: arrivals exceed service capacity, so backlog and priority order matter.
Landslide Corridor	Network reachability after closures: one bridge can isolate a region. Route planning: fewest roads and cheapest travel can disagree, so hop count cannot stand in for cost.
Outbreak Camp	Cargo selection: medicine/water urgency competes with expiry and limited capacity. Radio analysis: repeated beacon messages create highly repetitive input and deceptive near-miss patterns.